UNIVERSITI TEKNOLOGI MARA

TEST

NAME	:	
STUDENT NO.	:	
GROUP	:	
LECTURER	:	
DATE	:	
COURSE AND COURSE CODE	:	INTRODUCTION TO PROBABILITY AND STATISTICS (STA116)
TIME	:	2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of three (5) questions.
2. Answer ALL questions.
3. Do not refer any materials and make sure the answer script is completed before submit.

For Lecturer's Use Only:

QUESTION	MARKS
Q1 (10M)	
Q2 (10M)	
Q3 (10M)	
Q4 (10M)	
Q5 (10M)	
TOTAL (50)	

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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QUESTION 1

(a) Determine whether each of the following variables is qualitative, quantitative discrete, or quantitative continuous.

i) Top five bestselling car models in Malaysia. (1 mark)

ii) Income of the workers in a plastic manufacturing company. (1 mark)

iii) Time taken to serve a customer at the government clinic counter. (1 mark)

iv) Number of textbooks read in a year. (1 mark)

v) Brand of air conditioner. (1 mark)

(b) A finance executive from GDX Company is interested in conducting a study to estimate the monthly expenditure on transportation spent by the employees at GDX Company. This company has approximately 2400 employees. Each employee in the company was given a number and a sample of 400 employees is selected as respondents.

i. State the population and sample of the study.

(2 mark)

ii. Identify the variable of interest for this study and state its level of measurement.

(2 marks)

iii. What is the most suitable sampling method to be used?

(1 mark)

QUESTION 2

(a) The content of liquid detergent Y is being analysed. Twelve bottles were randomly selected from the production process and measured. The results are as follows (in litre).

1.53	1.57	1.61	1.57	1.49	1.51
1.52	1.62	1.49	1.53	1.51	1.59

Compute below for the data:

i. Mean and standard deviation. (3 marks)

ii. Median and interpret its meaning. (2 marks)

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(b) The following table shows the frequency distribution of commissions received by direct selling agents for a month in the Pasir Puteh area.

Commisions (RM)	No. of Agents
25 – 49	10
50 – 74	11
75 – 99	17
100 – 124	20
125 – 149	26
150 – 174	16

Draw a 'less than' ogive from the data above. (5 marks)

QUESTION 3

(a) How many different three-digit numbers can be formed using the digits 2,4,6,8,9 if no repetitions are allowed?

(2 marks)

(b) The word SYSTEMATIC is written on individual cards (one letter per card). All the cards are shuffled and placed in a row. Determine the number of ways the letters can be arranged.

(3 marks)

(c) In a seminar room, there are 15 men and 6 ladies. A consultant randomly selects 6 trainees present their ideas at the end of the seminar.

i. What is the probability that the group selected consists of at most **THREE (3)** men?

(3 marks)

ii. What is the probability that the group selected consists of all ladies?

(2 marks)

QUESTION 4

(a) A company decides to check on the accuracy of invoicing. Invoices are prepared by Alicia, Laura and Jane. A sample of 500 invoices are selected. Two hundred were prepared by Alicia, 120 by Laura and 180 by Jane. The error rates of Alicia, Laura and Jane were found to be 2%, 3% and 4% respectively.

i. Draw a tree diagram to represent this situation.

(3 marks)

ii. What is the probability that the invoice contains an error?

(2 marks)

iii. What is the probability that the invoice was prepared by Alicia, given that it contains no error?

(3 marks)

(b) Events A and B are such that $\mathrm{P(A)} = \frac{1}{8}$, $\mathrm{P(B)} = \frac{1}{2}$, and $\mathrm{P(A \cup B)} = \frac{1}{2}$. Find $\mathrm{P(A \cap B)}$.

(2 marks)

QUESTION 5

(a) In a box, there are 5 red pens, 7 blue pens, and 8 black pens. Two pens are drawn one after another without replacement.

i. What is the probability that both pens drawn are blue?

(2 marks)

ii. What is the probability that the first pen drawn is red and the second pen drawn is black?

(3 marks)

(b) Two events X and Y are defined in a sample space. P(X) = 0.28, P(Y) = 0.47 and $\mathrm{P(X \cap Y)} = 0.03$. Find:

i. $\mathrm{P(X \cup Y)}$

(2 marks)

ii. $\mathrm{P(X' \cap Y')}$.

(2 marks)

iii. If X and Y are independent.

(1 mark)

END OF QUESTION PAPER

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FORMULA LIST

Sample measurements

1. Mean, $\bar{x} = \frac{\sum x}{n}$ or $\frac{\sum fx}{n}$

2. Median, $\tilde{x} = L_m + \left(\frac{\frac{n}{2} - \sum f_{m-1}}{f_m} \right) \cdot C$

3. Mode, $\hat{x} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \cdot C$

4. Standard Deviation, $s = \sqrt{\frac{1}{n-1}\left[\sum (x - \bar{x})^2 \right]}$ or $\sqrt{\frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n} \right]}$

$\sqrt{\frac{1}{n-1}\left[\sum f(x - \bar{x})^2 \right]}$ or $\sqrt{\frac{1}{n-1}\left[\sum fx^2 - \frac{(\sum fx)^2}{n} \right]}$

5. Coefficient of variation, $cv = \frac{s}{\bar{x}} \times 100$

6. Pearson's measure of skewness = $\frac{\text{mean} - \text{mode}}{\text{standard deviation}}$ or $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$

where

n	=	$\sum f$
L_m	=	lower median class boundary
L_{mo}	=	lower modal class boundary
$\sum f_{m-1}$	=	cumulative frequencies for the classes before the median class
f_m	=	median class frequency
Δ_1	=	(modal class frequency) - (frequency for the class before the modal class)
Δ_2	=	(modal class frequency) - (frequency for the class after the modal class)
C	=	class size

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Probability

If A and B are two events,

1. Additive rule 1 (not mutually exclusive events)
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2. Additive rule 2 (mutually exclusive events)
$$P(A \cup B) = P(A) + P(B)$$

3. Multiplication rule 1 (independent events)
$$P(A \cap B) = P(A) \cdot P(B)$$

4. Multiplication rule 2 (dependent events)
$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

5. Conditional probability
$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

6. Bayes' Theorem
$$P(B_i \mid A) = \frac{P(B_i \cap A)}{P(A)} \quad \text{for } i = 1,2,...,k$$
where $P(A) = P(B_1) \cdot P(A \mid B_1) + P(B_2) \cdot P(A \mid B_2) + ... + P(B_k) \cdot P(A \mid B_k)$

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SOLUTION:

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